

ABIRAMI BASKARAN

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EDUCATION

Stanford University <i>International Honors Program Summer Session</i> Coursework: Design & Analysis of Algorithm and Energy Storage Integration	Jun 2026 - Aug 2026
National University of Singapore <i>Bachelor of Engineering, Electrical Engineering with Minor in Computing</i> Coursework: Digital Design, FPGA, Machine Learning, Data Structure Algorithm, Electrical Energy Systems, ...	Aug 2024 - May 2027
Singapore Polytechnic <i>Diploma in Computer Engineering, Specialised in Computer Application</i> Coursework: DevOps, OOP, DSA, MLAI, Client Server & Mobile Application, Embedded Computer, Microprocessor, ...	Apr 2022 - Mar 2024
Institute of Technical Education College Central <i>Higher Nitec in Electronics Engineering</i> Coursework: Digital & Analogue Principles & Applications, Electrotechnology, Advanced Applied Electronics, ...	Jan 2020 – Mar 2022

WORK EXPERIENCE

Defence Science Technology Agency (DSTA) – Land Systems <i>Robotics Engineer Intern (ROS2, Radio Communication, wireless video streaming, 3D modelling)</i> - Independently worked on big project with national-level contribution (NDP 2025), - Experience includes, integrating mechanical design, 3D modelling, and software development with user-friendly updates. - Independently implemented wireless video transmission across both local network and radio communication for reliable real-time streaming. - Gained hands-on experience through self-learning experimentations, combined robotics, communication systems, & mechanical concepts effectively	Jun 2025 – Aug 2025 Singapore, SG
Singapore Polytechnic – Digital Twinning Systems <i>Research Assistant (Power engineering, Machine Learning Algorithm, Artificial Intelligence, Digital Twin)</i> - Spearheaded the development of a digital twin and self-correction algorithm for Brushless DC motors to enhance system accuracy and performance. - Conducted extensive research on motor behaviour and applied findings to create an integrated application for real-time monitoring and analysis. - Took on extended responsibilities by planning project schedules and onboarding new interns to support ongoing research activities.	Mar 2024 – Jun 2024 Singapore, SG
Singapore Polytechnic – Digital Twinning Systems <i>Software & AI Engineer Intern (Digital Twin, Electric Vehicle Batteries, Neural Network, Web development)</i> - Developed a Digital Twin-based Battery Management System (BMS) for Electric Vehicle batteries, integrating AI with power engineering concepts to enhance fault detection and monitoring. - Designed and implemented algorithms that improved fault detection accuracy from 87% to nearly 100%, enabling more reliable system diagnostics. - Self-learned power technology fundamentals to bridge computing and electrical engineering, creating a 3D visual application for real-time battery performance tracking.	Sep 2023 – Feb 2024 Singapore, SG
IT Block – Tech Support <i>IT Technician Intern (IT Hardware & Software Support, Research Skills)</i> - Provided IT support and troubleshooting services to assist clients in resolving technical issues efficiently. - Conducted research and critical analysis to identify and implement effective solutions for system improvements. - Submitted weekly security reports and coordinated product deliveries, ensuring smooth operations and client satisfaction.	Jun 2021 – Aug 2021 Singapore, SG

PROJECTS & ACHIEVEMENTS

Best Freshmen Hack in Hack&Roll (24-hour hackathon) Built a robotic companion with image and speech recognition to assist users in exercising by following their actions and providing motivational feedback.	2025
First Prize Winner in NTU Deep Learning Week (4 days hackathon) Developed an application employing machine learning and computer vision to detect presentation habits, predict emotions through voice modulation, and deliver personalised feedback with generative AI.	2024
Championship in Dell InnovateFest (Inaugural Competition) Developed a Mobile Application with Augmented Reality, real-time chat, and a unique interface catered towards people with intellectual disability (PWID). With support from MINDS, the app was suitable for use and encourages PWID to socialise and play sports.	2023
People's Choice Award in Samsung Solve For Tomorrow (Top 5 Finalist) Built a well-rounded robotic bear controlled by a mobile app for paediatric children. With a pulse and thermal sensor nurses can monitor and with a real time voice communication, parents can speak with their child while working.	2022

Additional projects available at: [portfolio website](#)

LEADERSHIP

NUS ECE Undergraduate Student Council – President Overseeing a team to plan and execute academic, social, and community initiatives for over 1,000 students.	Aug 2025 – Present
Project Director: NUS ECE Internship & Career Fair' 2026 - Engaged 500+ students through targeted outreach and on-ground coordination, driving strong participation and interaction - Secured and managed partnerships with 28 companies, establishing structured and professional industry engagement - Led a cross-functional organizing team, overseeing planning, delegation, and execution of a large-scale faculty event	28TH January 2026

TECHNICAL SKILLS AND TOOLS

Programming languages/frameworks: Python, Java, PHP, Assembly, Verilog, C++
Technical Skills: Data Structure Algorithm, Object Oriented Programming, Web & Mobile Development, Artificial Intelligence, IOT, Microcontroller Applications, Embedded Systems, Computer Interfacing, Data Engineering, FPG System
Tools: Pandas, NumPy, Flask, MongoDB, Computer vision, Autodesk Fusion 360
Certifications: IOT, Google IT Support, Docker Fundamentals, Kubernetes Fundamentals